

Can squirrel behaviour patterns predict whether a squirrel will approach humans in Central Park?

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Executive Summary

In this study we use the 2018 squirrel census data collected in New York's Central Park to explore our problem. The dataset contains roughly 3,000 entries, covering detailed records of squirrels' information.

Preprocessing dealt with missing values, removed columns with high missing rates, and merged squirrel and hectare-level data. Because the dataset is imbalanced, only about 6% of squirrels approaching humans, random oversampling was applied to the training set.

Mutual Information and Pearson correlation analyses showed that "Indifferent" is the strongest behavioural predictor of "Approaches", food related behaviours such as Eating and Foraging also show clear associations.

KNN and Decision Tree models training applied oversampling to address imbalanced data, then testing both models on raw authentic datasets separately, they all show high accuracy rates. However, because of the highly imbalanced, greater emphasis should be placed on recall and F1 score.

Cluster analysis from three perspectives, behavioural traits, geographical locations and risk warning behaviours. Results revealed that squirrels which tend to forage frequently, inhabit southern areas and show no tail-flicking alert movements are most likely to approach humans.

In conclusion, squirrels' willingness to get close to humans is determined by multiple influencing factors rather than a single one.

Introduction

Urban green spaces serve as everyday shared habitats where people and wildlife regularly come into contact. As one species well-adapted to living alongside humans, the eastern grey squirrel is frequently spotted in urban parks across North America. Unlike most urban wildlife that tend to steer clear of human activity, squirrels display highly varied reactions to people, from fleeing or sounding alarm calls to actively moving toward park-goers. This broad spectrum of behavioural responses makes eastern grey squirrels an ideal subject for exploring how distinct behavioural traits shape human-wildlife interactions within city environments.

The central research question:

“Can squirrel behaviour patterns predict whether a squirrel will approach humans in Central Park?”

The 2018 Central Park Squirrel Census, a systematic dataset that records detailed behavioural, spatial, and contextual information of squirrels in Central Park, New York, also provides structured large-scale observations of individual squirrels collected during October 2018 (2018 Central Park Squirrel Census - Squirrel Data | NYC Open Data, 2019).

The dataset analysed in this study originates from two core files collected in the New York City Squirrel Census. The `squirrel.csv` file records individual-level squirrel observations, including behavioural patterns, vocal and signalling activities, physical traits, geographic locations, and crucially, whether each squirrel approached human observers. The `hectare.csv` file supplies contextual environmental data at the hectare scale across the park, such as squirrel population density per survey shift, weather conditions and time-related observation metadata. We combined these two datasets for integrated analysis. A third supplementary file, `stories.csv`, was omitted from our modelling due to excessive missing values and its irrelevance to our predictive objectives.

The analysis is structured across four methodological components: preprocessing, correlation analysis, supervised learning, and clustering. Each chosen to address a different aspect of the research question from complementary perspectives. Together, they allow both predictive performance and underlying behavioural structure to be examined in relation to approach behaviour.

Methodology

2.1 Data Preprocessing

The study mainly used two datasets: `squirrel.csv` and `hectare.csv`. When it came to preprocessing, we basically followed three ideas — get rid of columns that were mostly empty, fill in missing values in a way that still made sense, and keep anything that might actually be useful for prediction.

For `squirrel.csv`, we dropped eight columns where more than 84% of the data was missing. These included things like color notes, specific location, other activities, other interactions, and some ID columns that were basically duplicates anyway. We did the same for `hectare.csv`, removing three columns for the same reason. Since the majority of columns in the `stories.csv` dataset contained 50% to 97% missing entries, and the available data mainly consisted of unstructured free-text notes with little predictive value, we decided not to use it at all.

For the columns we kept, we tried to fill in missing values without distorting the original meaning too much. In the age column, entries that were blank or marked as "?" got replaced with "Adult", since that was the most common value. We did something similar for primary fur color and location, both filled using their mode. For highlight fur color specifically, we used "None" instead of a mode, because in this case a missing value most likely just means the squirrel didn't have any highlighted fur.

The above-ground sighting height column needed a bit more thought. There were entries labelled "FALSE" which actually meant the squirrel was spotted at ground level, so we converted these to 0 and treated the whole column as a continuous numerical feature. We did consider making it a binary categorical variable at first, but decided against it — keeping it numerical felt more intuitive since the values actually represent real height levels.

The two datasets were then merged using a left join, matched on hectare area, time shift, and date. We additionally constructed one new feature that squirrel count per hectare to represent local population density within each park zone.

Following all data-cleaning procedures, our final dataset contained 3,023 samples across 24 features, with no remaining missing values.

Before we started using oversampling methods, we split our dataset into a training set (80%) and a test set (20%), and we used `random_state=42` for this split. We made sure to do the split first on purpose—this way, the test set could keep the actual class distribution of the real world and wouldn't be influenced by the oversampling process. If we had done these steps in the wrong order, our evaluation results wouldn't have been accurate, and they wouldn't really show how the model would work when it's used on new, unseen data.

2.2 Correlation Analysis

In this section, we use two complementary methods to analyse the relationship between features and the target variable, Mutual Information (MI) and Pearson correlation. We use these two together because they capture different aspects of the data, which can provide a more complete understanding and deeper analysis.

Mutual Information is used as the main feature measure method in this study. Because it can quantify the level of association between each feature and the target variable, without assuming a linear relationship. Unlike Pearson correlation, which only detects linear relationships, Mutual Information can also capture non-linear patterns. This makes it more suitable for this dataset, which includes both binary variables (such as whether certain behaviours occur) and categorical variables (such as fur colour, age, location, and time of observation).

In the calculation of Mutual Information, all features were included. These consist of boolean behavioural variables, categorical variables (Age, Primary Fur Color, Location, Shift), and continuous numerical variables (such as Above Ground Sighter Measurement and Hectare_Squirrel_Count). Boolean and categorical variables were treated as discrete, while numerical variables were treated as continuous, which is consistent with the `'mutual_info_classif'` function in scikit-learn. In addition, label encoding was used only to convert categorical variables into a numerical format for computation, encoded values do not represent any order or magnitude relationship.

Pearson correlation analysis was used as a supplementary method in this report, to examine the linear relationships between the 12 boolean behavioural features and the target variable, "Approaches". Although Pearson correlation assumes linear relationships, is not the most ideal method for binary variables, it still provides an advantage that Mutual Information does not have, that it shows the direction of the relationship, indicating whether a feature is positively or negatively associated with approach behaviour. Therefore, it was mainly used to complement the Mutual Information results and to examine whether a behaviour increases or decreases the likelihood of approaching humans.

In addition, regarding the selection of those methods, Spearman correlation was considered as an alternative to Pearson. Since it is more stable for non-normality and better suited to ordinal or skewed distributions. However, in this project, all features in the heatmap are binary (0/1), which means the Pearson and Spearman coefficients produce near-identical results for such variables, so we think Spearman correlation analysis was unnecessary. Point-biserial correlation was also considered for

examining binary-to-binary relationships, but as this is mathematically equivalent to Pearson correlation when both variables are binary, no additional analysis was required.

2.3 Supervised Learning

Classifiers such as Decision Tree and K-Nearest Neighbours (KNN) are implemented in this task. KNN classifies through the similarities of the adjacent samples and requires no data distribution assumption. Decision Tree's interpretability is based on tree structure, directly showing which features and thresholds drive the predictions. When using both of the classifiers, they provide complementary perspectives. More complex models such as Random Forest or SVM were not used to maintain interpretability, though these are identified as alternatives that could improve F1 by capturing feature interactions.

Class imbalance (5.9% True) was addressed by applying RandomOverSampler exclusively to the training set (2,418 samples to 4,542 after oversampling). SMOTE, which generates synthetic minority samples by interpolation, was considered as an alternative but not implemented to avoid introducing synthetic points that may not reflect real squirrel behaviour patterns. Cost-sensitive learning was also considered but not used to preserve comparability between the two models.

For KNN, StandardScaler was applied to normalise features, fitted on the oversampled training set and applied to the test set to prevent leakage. No scaling was applied for the Decision Tree, as tree splits are invariant to feature scale. `max_depth=4` was set after observing that deeper trees showed training accuracy near 1.0 while test performance degraded, confirming overfitting.

Models were evaluated using four metrics: accuracy, precision, recall, and F1 score. Accuracy alone is misleading under class imbalance, as a baseline model that always predicts the majority class achieves 94.1% accuracy while correctly identifying zero positive cases. Precision and recall capture the reliability and completeness of minority class predictions respectively, and F1 score provides a single balanced measure that accounts for both.

2.4 Clustering

In this part, we conduct three clustering analyses on distinct feature subsets to address different dimensions of the research question.

The first clustering analysis is behavioural and temporal clustering. We use the K-Means method on activity features (Running, Chasing, Climbing, Eating, Foraging) with Shift (AM=1/PM=0). The reason we use K-Means is that the features are binary or binary-encoded and have the same scale, making Euclidean distance appropriate without normalisation. We use the Elbow method over $K = 1 - 10$ and use the Silhouette Scores for $k = 3$ and 5 then choose the high scores when $k = 5$. Also we choose K-Means here instead of hierarchical clustering because the flat partition structure is easier to represent and compare the data against Approaches.

The second Geospatial Clustering is still K-Means on coordinates (X,Y). Due to the same unit, we don't need the normalisation. We use the Elbow method again and use the Silhouette Scores to compare which K value is more appropriate and finally choose $K = 2$ to divide it into two parts (south part and north part).

The third clustering we use is agglomerative hierarchy clustering to analyze the signal feature (Kuks, Quaas, Tail flags, Tail twitches). Compare ward linkage with single linkage(sensitive to outliers, produces chaining) and complete linkage(can split natural groupings), ward linkage can minimise the within-cluster variance, producing compact and better clusters. So $K = 3$ was selected and used to do the dendrogram.

Results Exploration and Analysis

3.1 Preprocessing Results

After merging every data together, we had 3,023 samples and 24 features, and there were no missing values in the final dataset. Out of all the preprocessing steps we did, probably the most important one was how we handled the above-ground sighting height variable. Instead of treating "FALSE" as a category, we converted it to 0 and kept it as a numerical variable — we felt this made more sense because turning it into a dummy variable would've thrown away some of the information. This turned out to matter quite a bit, since this feature ended up having the second-highest MI score ($MI = 0.0128$) among all the variables we tested.

We also added a new feature for squirrel density (squirrels per hectare, $MI = 0.0017$). The idea was that a single behaviour variable probably can't capture the full picture of local population pressure, so we thought density might add something extra. Whether it actually helps the model much is another question, but we included it anyway.

It is also important to note the age variable, which contained 121 missing entries. We handled these missing values using mode imputation, meaning most unknown entries were replaced with the label "adult". This may have slightly biased our dataset toward adult squirrels. However, given that the mutual information score for age was only 0.0001, this imputation choice is unlikely to meaningfully impact our final model performance.

Table 1: Preprocessing Decisions and Their Impact on Downstream Analysis

Column	Issue	Treatment	Downstream Impact
Color notes, Specific Location, Other Activities, Other Interactions	>84% missing	Dropped	No signal lost
Age	121 missing + '?' values	Mode imputation (Adult)	$MI=0.0001$; negligible
Primary Fur Color	55 missing	Mode imputation (Gray)	$MI=0.0032$; minor
Highlight Fur Color	1,086 missing (36%)	Filled 'None'	Semantic imputation
Location	64 missing	Mode imputation	$MI=0.0047$; small
Above Ground SighterMeasurement	114 missing + 'FALSE'	FALSE→0, cast to float	2nd-highest MI (0.0128)
Hectare_Squirrel_Count	Left join from hectare.csv	Median fill	Density context $MI=0.0017$
stories.csv	50–97% missingness	Excluded entirely	No signal for Approaches

3.2 Correlation Analysis Results

As shown in Figure 1, features are ranked from highest to lowest MI score. Higher scores indicate stronger statistical dependency with the target. Indifferent (0.0201) and Above Ground Sighter Measurement (0.0128) are the two features most strongly associated with Approaches, substantially outranking all others. Location (0.0047), Primary Fur Color (0.0032), Runs from (0.0026), and Eating (0.0024) form a secondary tier, the moderate relevance. Features such as Shift, Tail flags, Running, and Kuks scored at or near zero, indicating negligible association with the target. Therefore, in the subsequent supervised learning stage, they were used as an important basis for feature elimination.

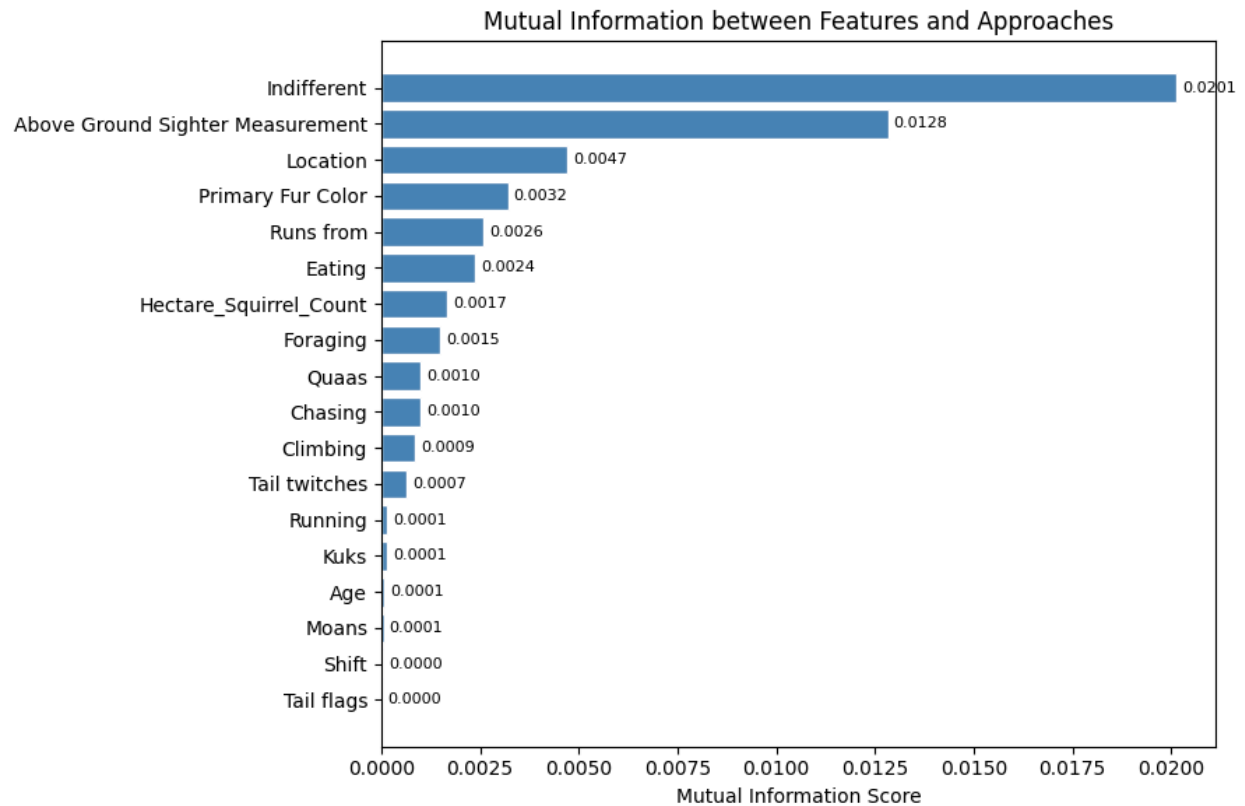


Figure 1: Mutual Information scores between all features and the target variable Approaches.

The mutual information (MI) results show that among all features, “indifference” is the strongest indicator of whether a squirrel will approach humans. In other words, whether a squirrel shows a lack of reaction toward humans is the most helpful signal for predicting if it will come closer. In addition, the “Above Ground Sighter Measurement” also provides some useful information. This may reflect the squirrel’s level of alertness or how comfortable it is in a human environment.

However, features such as whether the squirrel makes sounds or its movement state, have almost no relationship with approaching behaviour, which means they contribute very little to predicting whether a squirrel will approach humans.

Figure 2, the Pearson correlation heatmap, only the lower triangle is displayed to avoid redundancy. Colour intensity indicates the magnitude of correlation, red indicates positive and blue indicates negative association.

In the graph, we can see that “Indifferent” has a negative correlation with “Approaches” ($r = -0.19$), indicating that squirrels showing indifferent behaviour are less likely to approach humans. This directional finding is consistent with the Mutual Information results, but Mutual Information alone cannot provide information about positive or negative relationships.

“Runs from” shows a relatively strong negative correlation with “Indifferent” ($r = -0.47$), indicating that these two behavioural patterns are largely mutually exclusive responses. “Eating” ($r = 0.07$) and “Foraging” ($r = 0.05$) both show weak positive correlations with approach behaviour, supporting the idea that food-motivated squirrels are more likely to approach humans.

At the same time, “Foraging” is negatively correlated with both “Climbing” ($r = -0.30$) and “Running” ($r = -0.22$), suggesting that squirrels engaged in ground-level food searching are less likely to be in rapid movement states. This structural behavioural pattern appears consistently within the dataset.

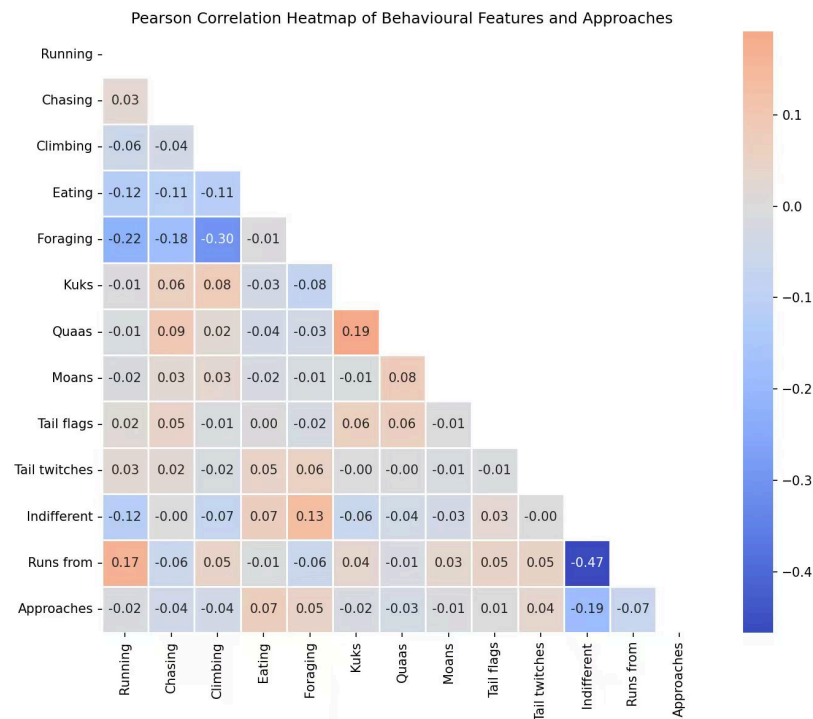


Figure 2: Pearson correlation heatmap of boolean behavioural features and Approaches.

Overall, both methods suggest that “Indifferent” is the most strongly associated behavioural feature related to whether squirrels approach humans. In addition, food-related behaviours, although not the strong predictors, still generally show a tendency toward a higher likelihood of approaching humans.

However, an important point is that the relationships between all features and the approach behaviour are actually quite weak. Even the most informative feature is not strongly predictive on its own. This indicates that no single behaviour can reliably predict whether a squirrel will approach humans.

Therefore, in the next section (3.3), a multivariate machine learning model is needed, as such methods can consider the combined effects of multiple features rather than relying only on simple relationships between individual variables.

3.3 Supervised Learning Results

Figure 3 confirms severe class imbalance of 2,845 squirrels (94.1%) do not approach humans, while only 178 (5.9%) do. This imbalance is the central challenge for all models, where a naive classifier always predicting False would achieve 94.1% accuracy while identifying zero true approach cases, making accuracy alone a misleading evaluation metric. Table 3 presents performance across four metrics for all models on the held-out test set (605 samples).

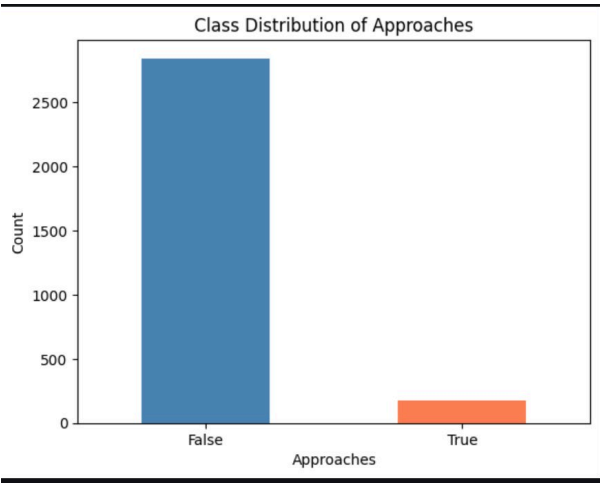


Figure 3: Class Distribution of Approaches — 2,845 False (94.1%) vs 178 True (5.9%)

Table 3: Model Performance on Test Set (605 samples; 5.9% True class)

Model	Accuracy	Precision	Recall	F1 Score
Majority Class Baseline	0.94	—	0.00	0.00
KNN (k=5)	0.94	0.31	0.13	0.18
Decision Tree (max_depth=4)	0.68	0.12	0.77	0.20

The majority-class baseline confirms that 94% of accuracy with zero recall and zero F1, demonstrating that high accuracy does not imply useful predictions under class imbalance. Both trained models must be evaluated against this baseline to assess whether they are genuinely learning from the data.

Figures 4 and 5 show the confusion matrices for KNN and the Decision Tree respectively. KNN matched the baseline accuracy (0.94) with only 4 true positives and 27 false negatives. Its precision of 0.31 indicates that when KNN does predict approach, it is correct 31% of the time, but its recall of 0.13 means it missed 87% of squirrels that actually approach. Despite oversampling, KNN's local neighbourhoods in the test space remain dominated by False samples, causing systematic bias toward the majority class.

Oversampling helped the training distribution but could not overcome the fundamental imbalance at test time.

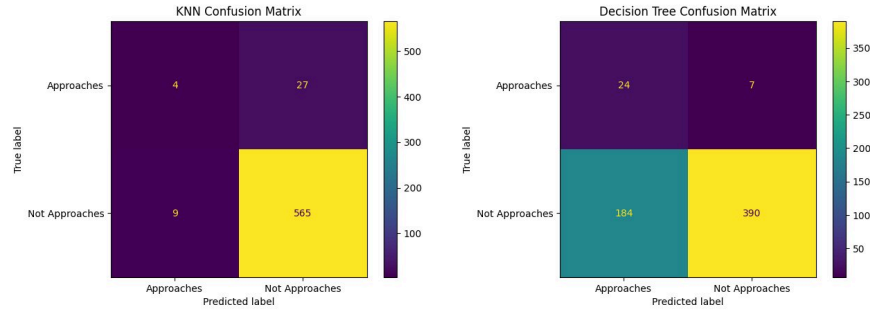


Figure 4 (left): KNN Confusion Matrix **Figure 5 (right): Decision Tree Confusion Matrix**

The Decision Tree shows the opposite trade-off of a lower accuracy (0.68) but a recall of 0.77, successfully identifying 24 of 31 true approach cases with only 7 false negatives, at the cost of 184 false positives. To address the research question (which squirrels will approach humans), the Decision Tree is a more useful model. Missing a genuine approach case is more informative to avoid than generating a false alarm. The similar F1 scores (KNN: 0.18, DT: 0.20) mask fundamentally different error profiles: KNN minimises false positives at the expense of recall, while the Decision Tree prioritises recall at the expense of precision.

Figure 6 visualises the Decision Tree structure. The root split on Indifferent confirms the MI findings from Section 3.2, as a squirrel's disposition toward humans is the single most discriminative feature. The second-level splits on Runs from and Foraging introduce food-seeking context, consistent with their MI scores and feature importances in Table 4.

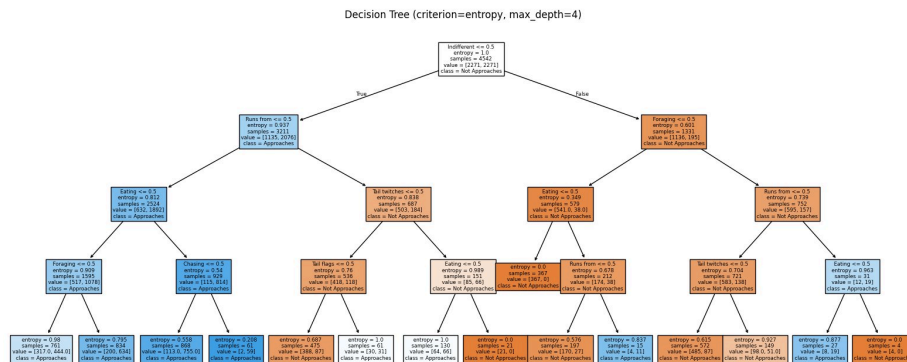


Figure 6: Decision Tree Structure (criterion=entropy, max_depth=4) — Root split on Indifferent

The consistent results obtained from two independent approaches, namely information-theoretic mutual information and tree-based splitting rules, further verify that the derived feature importance rankings truly reflect the inherent behavioral patterns of squirrels instead of spurious outcomes generated by a single modeling method.

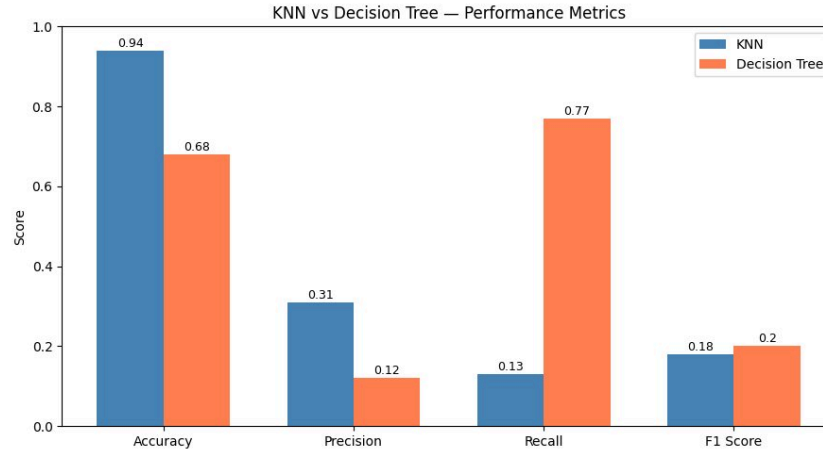


Figure 7: KNN vs Decision Tree — All Performance Metrics Side by Side

Table 4: Decision Tree Feature Importances vs. Mutual Information Scores

Feature	DT Importance	MI Score	Role in Tree
Indifferent	0.495	0.0201	Root split
Runs from	0.284	0.0026	2nd-level split (left branch)
Eating	0.123	0.0024	3rd-level split
Foraging	0.056	0.0015	2nd-level split (right branch)
Tail twitches	0.025	0.0007	4th-level split
Tail flags	0.014	0.0000	Minor leaf splits
Running / Climbing / Kuks / Moans / Quaas	0.000	≤ 0.0001	Not used

Table 4 shows that both models yield persistently low F1 scores, which indicates the inherent difficulty in predicting approaching behaviour rather than defects in model selection. As mentioned in Section 3.2, all mutual information values are below 0.021, suggesting that no single behavioural feature carries sufficient independent predictive information. This finding aligns with the class imbalance of the dataset, in which most squirrels tend to avoid humans, leading both models to be biased towards predicting the majority class. Features including running, climbing, moaning and quacking lack discriminative capacity and are thus completely eliminated in the tree model.

3.4 Clustering Results

Three independent clustering analyses were conducted and each addressing a different dimension of the research question.

For the first one: Behavioural and Temporal Clustering (K-Means and K=5). Figure 8 shows the Elbow Method; Figure 9 shows the feature means heatmap for K=5.

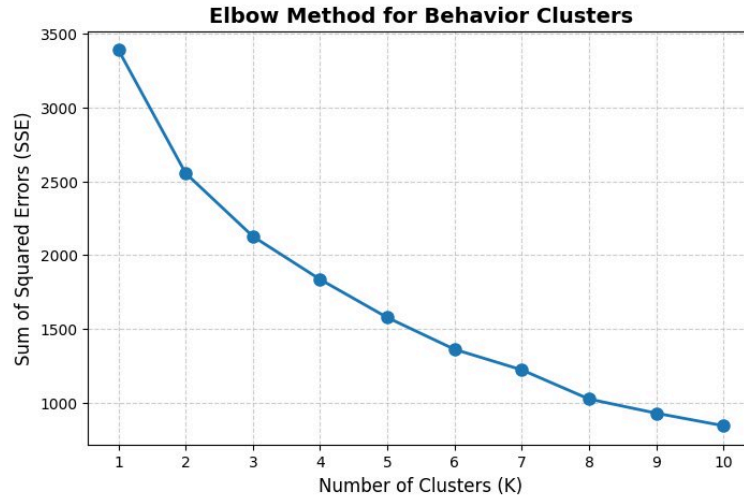


Figure 8: Elbow Method for Behavioural Clusters — Elbow visible around K=4–5

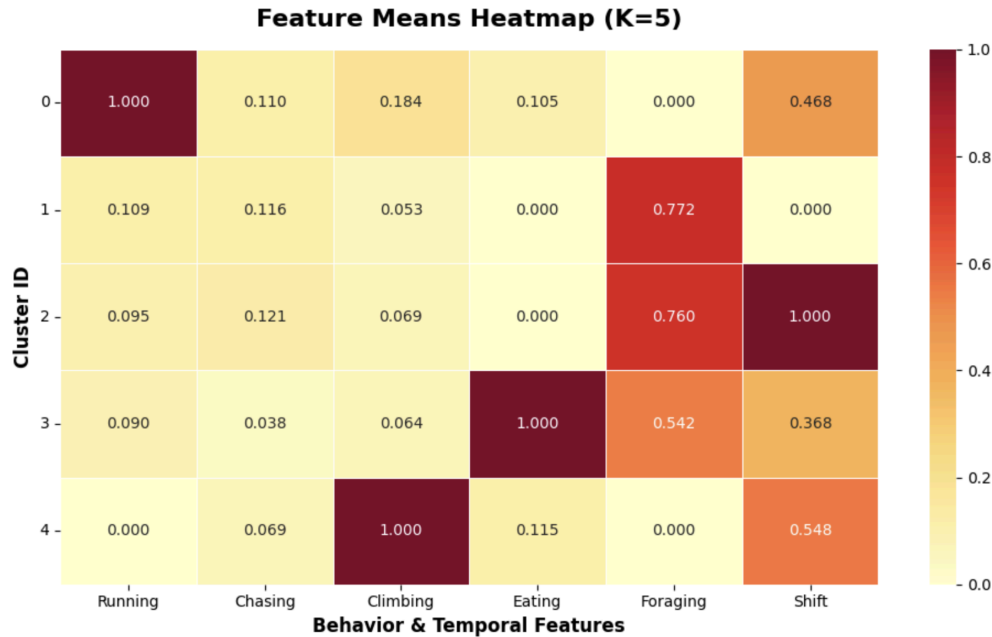


Figure 9: Feature Means Heatmap (K=5) — Five distinct behavioural profiles

The Elbow Method started to level off around $K = 4-5$; but $K = 5$ has a higher Silhouette Score than $K = 3$. The heatmap shows five interpretable profiles. There are obvious features in each cluster, cluster 0 is mainly running, cluster 1 is foraging in PM, cluster 2 is foraging in AM, cluster 3 is eating and cluster 4 is climbing. Compare that with approach rate we found that: Cluster 4 (Climbers) shows the lowest approach rate which is just 3.2%, logically similar with their physical distance from trees to peoples. Conversely, Cluster 3 which are eating(1.0) and foraging (0.54) showed the highest approach rate at 8.5%. It indicates that squirrels actively engaged in food-seeking behaviors are the most likely to view humans as potential food sources and approach them.

Geospatial Territory Segmentation (K-Means, $K=2$). Figure 10 shows the Elbow Method; Figure 11 shows the spatial map.

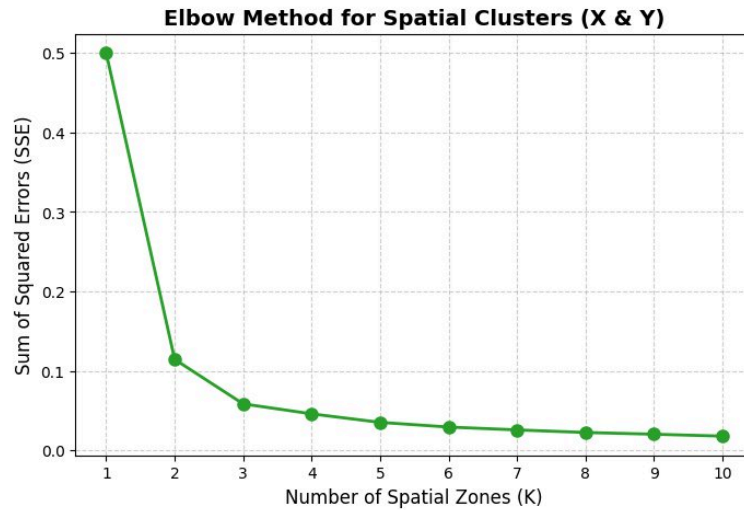


Figure 10: Elbow Method for Spatial Clusters — Sharp elbow at K=2

Spatial Distribution of Squirrels in Central Park (K=2)

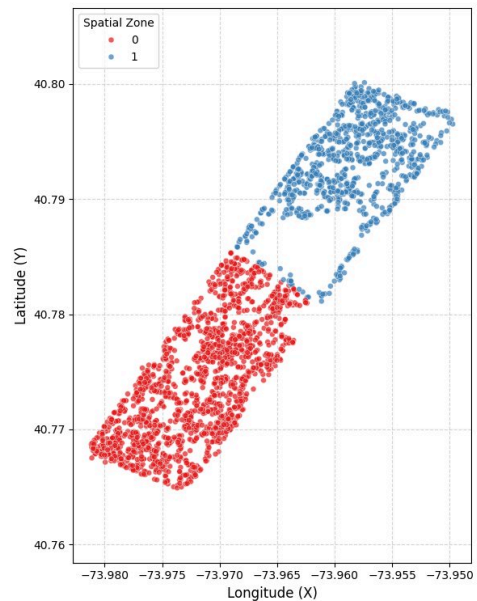


Figure 11: Spatial Distribution of Squirrels in Central Park (K=2) — Red = Southern Zone, Blue = Northern Zone

K=2 has a higher Silhouette Score than K=3 and a clear north-south division. Southern area squirrels (Cluster 0) approach at 6.4% versus 4.9% in the Northern Zone (Cluster 1). The difference is analytically meaningful: the Southern Zone is near 59th Street, the Central Park Zoo, and the MET which are higher tourist density. So squirrels in this area likely exhibit greater human habituation and food-seeking behavior. In contrast, the quieter, heavily wooded Northern Zone fosters behavior that is relatively more wild which means that geographical location is a valid macro-predictor of squirrel-human interaction

Warning Signal Hierarchy Analysis (Agglomerative Hierarchical Clustering with ward linkage, K = 3). Figure 12 shows the dendrogram.

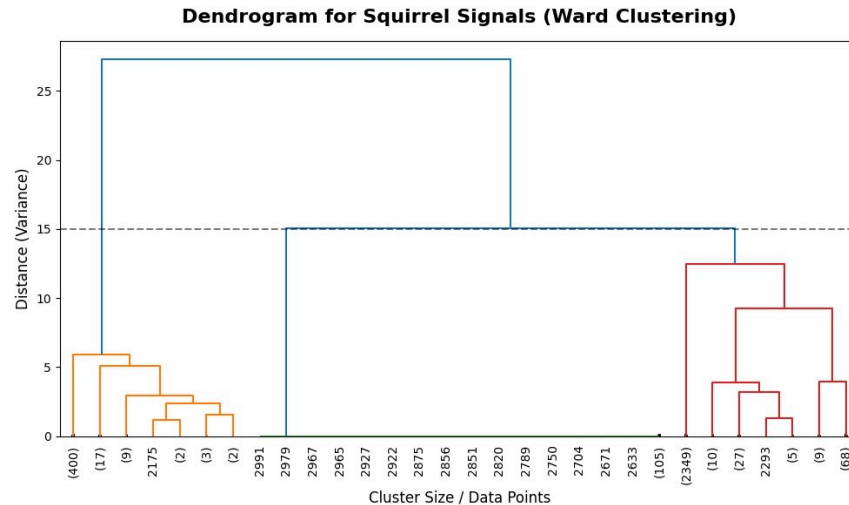


Figure 12: Dendrogram for Squirrel Warning Signals (Ward Clustering) — Dashed line at distance=15 indicates K=3 cut

Three clusters emerged: Cluster 0 (unexpressive, approach rate 5.5%), Cluster 1 (Tail twitches only, approach rate 8.0%), and Cluster 2 (Tail flags only, approach rate 6.6%). Cluster 1 ('Tail twitches') has the highest approach rate. On the other hand, the quiet group (Cluster 0) has the lowest rate. It makes sense because tail twitching usually indicates curiosity or observation. Therefore, these squirrels are naturally more open to interacting with people compared to the completely unresponsive ones.

Discussion and Interpretation

The following findings emerged consistently across all four analytical stages, each individually supported by multiple methods:

Indifference and fleeing behaviour appear to be key factors predicting whether squirrels will approach humans. Of these two traits, indifference yielded the highest mutual information score (0.0201) and the greatest feature importance value (0.495) in our decision-tree model, and was used for root-node splitting. There is a significant negative Pearson correlation between indifference and fleeing ($r = -0.47$), supporting the view that individual squirrels are either habituated to human presence or fearful of humans. Importantly, this correlation does not imply causation; both behaviours are likely shaped by unmeasured background factors, such as long-term habituation levels and past experiences of receiving food rewards from humans. The squirrel's vertical position when observed (Above Ground Sighter Measurement) had the second-highest mutual information score (0.0128), noticeably higher than all other behavioural variables. This suggests that height provides unique predictive information on squirrel-human interactions. Squirrels observed on the ground were more inclined to interact with humans, yet this trend is merely correlational instead of causal. While it may seem that location directly shapes behaviour, differences in prior human exposure are more probably driving these behavioural differences.

Food-seeking behavior serves as the second major factor driving approaching behavior. Eating and foraging display steady importance across various approaches, including mutual information values (0.0024, 0.0015), decision tree feature importance (0.123, 0.056), and clustering outcomes, in which the food-seeking group achieves the highest approach rate at 8.5%. The consistent findings across diverse methods indicate that squirrels actively foraging are more prone to regard humans as potential food sources.

Geographic location modulates approach rates at a macro level. The Southern Zone's higher approach rate (6.4% vs 4.9%) reflects long-term habituation from tourist activity near high-footfall areas. This effect is invisible to the supervised models, which do not include spatial features, suggesting that incorporating location or tourist-density features could meaningfully improve classification performance.

The precision-recall trade-off is the central modelling challenge. KNN's 94% accuracy matching the majority-class baseline demonstrates that accuracy is a misleading metric under class imbalance. The Decision Tree's recall of 0.77 is more relevant to the research question: identifying true approach cases matters more than avoiding false alarms. The similar F1 scores (0.18 vs 0.20) mask fundamentally different error profiles that must be evaluated in context. KNN identified only 4 of 31 true approach cases, while the Decision Tree identified 24 of 31, which shows a difference that is invisible if accuracy alone is reported.

Clustering results provide structural validation for the supervised learning findings. The convergence across behavioural, geographic and signal clustering analysis analyses all identify the same approach-prone profiles. It strengthens confidence that these patterns reflect genuine data structure rather than modelling artefacts.

Also, curious signals like tail twitches are a better sign of a squirrel approaching than defensive tail flags. The cluster with only tail twitches had an 8.0% approach rate, while the quiet squirrels were at 5.5%. This demonstrates the value of clustering over pairwise correlation when group observations reveal patterns that individual associations cannot capture.

All associations are non-causal. The features reflect observational snapshots; confounders such as proximity to food carts, individual habituation history, and time since last human contact are absent from the dataset and likely drive the underlying behaviour.

Limitations and Improvement Opportunities

Several specific limitations affect the strength of conclusions that can be drawn.

The most fundamental limitation is severe class imbalance (5.9% True). RandomOverSampler duplicates existing observations rather than generating new ones, which may cause models to overfit to the specific 178 True examples. SMOTE, which interpolates between minority-class points, could improve generalisation. Cost-sensitive learning explicitly penalising false negatives during training, offers an alternative that avoids resampling entirely and could be implemented without changing the model architecture.

All features are point-in-time binary snapshots. Approach behaviour is influenced by contextual factors absent from the dataset: whether a human nearby is offering food, individual habituation history, or time since last contact. These unmeasured confounders place a ceiling on predictive performance that no modelling technique can overcome with the current feature set. Augmenting the dataset with proximity-to-food-source estimates, tourist density measures, or temporal features would be the most productive improvement.

The mutual information (MI) scores for continuous features are estimated using a k-nearest neighbours approach. This method involves a certain degree of randomness, so the results may be slightly different from each run. For features in the middle range, such as Location, Runs from, and Eating, their scores are very close to each other. As a result, their relative rankings may change under different random seeds. Therefore, these MI results should be a general reference rather than a strict or definitive ranking.

The limitation of Pearson correlation is that it only captures linear relationships between two variables, meaning it examines the simple relationship between one factor and the target at a time. However, it cannot capture the combined effects of multiple features. For example, when “Indifferent = True” and “Eating = True” occur together, they may provide stronger evidence for whether a squirrel will approach humans than either feature alone. However, this kind of interaction effect cannot be detected by Pearson correlation.

The Decision Tree captures this partially through its hierarchical split structure, but a Random Forest or gradient-boosted tree ensemble would capture higher-order interactions more systematically while maintaining reasonable interpretability.

Clustering results are sensitive to feature subset, distance metric, and K value. The Elbow Method and Silhouette Score provide guidance but do not uniquely determine K. The spatial K=2 partition simplifies what is likely a more complex gradient of human exposure across the park. The signal hierarchy clustering with Ward linkage makes clean separations that other linkage methods would not, which could be interpreted as a method-induced artefact rather than a true data structure.

Conclusion

This study focuses on whether squirrel behavioural patterns can predict approach behaviour toward humans in Central Park. The result shows that the behavioural patterns of squirrels have a certain predictive ability for their behaviour when approaching humans, but this predictive ability is limited and is constrained by multiple factors.

Across all the methods we tried, behaviours where squirrels acted indifferent or ran away were the most consistent predictors. Actions linked to feeding and foraging also provided small hints, but the impact is relatively limited. Furthermore, the location of the squirrels within the park seems to independently affect the approach rate beyond their behavioural characteristics, which suggests that there may be some environmental factors.

For the model selection, we compared two models’ performance, decision trees worked noticeably better than K-NN. Its recall rate is 0.77 and the F1 score is 0.20. It successfully identified most of the actual true approach cases. On the other hand, K-NN failed to meaningfully outperform the majority-class baseline despite oversampling. KNN predictions are dominated by the majority class.

Overall, whether a squirrel approaches humans does not depend on one single factor. It comes from the result of the interaction between the squirrel’s inherent nature and the surrounding environment. One of the biggest limits to our prediction power is missing background information, such as each squirrel’s past habits, how far they are from food, and specific interaction situations. Simply increasing the complexity of the model cannot fundamentally overcome this deficiency. Therefore, if future research aims to make substantive progress on this basis, it is necessary to systematically incorporate more comprehensive contextual features during the data collection stage. This would help us better understand how human-squirrel interactions happen across Central Park.

References

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